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1924 Alcoa Highway
Knoxville, TN 37920

Surgical Pathology Report

Patient:	**HILL, LESLIE A**	Accession #:	**S23-1449**
Med. Rec. #:	2141502	Client:	0000
DOB/Gender:	11/26/1971 (Age: 51) F	Fin #:	21415020009
Soc. Sec. #:	xxx-xx-0692	Taken:	1/18/2023
Location:	Day Surgery	Received:	1/18/2023
Physician(s):	Matthew L. Mancini, M.D.	Reported:	1/26/2023

Specimen(s) Received

A:Sentinel lymph node #1
B:Possible sentinel lymph node #2
C:Nonsentinel lymph node
D:Left hookwire partial mastectomy
E:Deep margin

Clinical History and Diagnosis

Breast cancer

Final Pathologic Diagnosis

A) LYMPH NODE, SENTINEL #1, LEFT AXILLA, EXCISION:
NEGATIVE FOR MALIGNANCY AND ISOLATED TUMOR CELLS IN ONE LYMPH NODE

B) "LYMPH NODE, POSSIBLE SENTINEL #2", LEFT AXILLA, EXCISION:
1) BENIGN FIBROADIPOSE TISSUE
2) LYMPHOID TISSUE IS NOT PRESENT

C) "LYMPH NODE, NON-SENTINEL", EXCISION:
1) BENIGN FIBROADIPOSE TISSUE
2) LYMPHOID TISSUE IS NOT PRESENT

D) BREAST, LEFT, HOOKWIRE PARTIAL MASTECTOMY:
1) INVASIVE MAMMARY CARCINOMA COMPATIBLE WITH MUCINOUS TYPE
HISTOLOGIC GRADE: GRADE 3, POORLY DIFFERENTIATED
NOTTINGHAM SCORE: 8 OF 9 (TUBULES - 3, NUCLEI - 3, MITOSIS - 2)
SIZE: 16.0 MM
ANGIOLYMPHATIC INVASION: NOT IDENTIFIED
PERINEURAL INVASION: NOT IDENTIFIED
MARGINS: FREE OF INVASIVE TUMOR, CLOSEST MARGIN IS POSTERIOR AT 2.0 MM
ANCILLARY STUDIES: PERFORMED BY OUTSIDE INSTITUTION, SEE SYNOPTIC SUMMARY
2) DUCTAL CARCINOMA IN SITU (DCIS)

PATTERN(S): SOLID, COMEDO
SIZE: SPANS APPROXIMATELY 40.0 MM
NUCLEAR GRADE: 3
MARGIN: NARROWLY FREE OF DCIS (DCIS IS LESS THAN 1.0 MM FROM POSTERIOR, SUPERIOR AND INFERIOR MARGINS; DCIS IS 1.0 MM FROM THE ANTERIOR MARGIN; MEDIAL AND LATERAL MARGINS ARE WIDELY FREE OF DCIS; POSTERIOR MARGIN RE-EXCISED AS SPECIMEN E

3) ADDITIONAL FINDINGS:
LOBULAR CARCINOMA IN SITU
FIBROCYSTIC CHANGE
USUAL DUCAL HYPERPLASIA (UDH)
INTRADUCTAL PAPILLOMA
BIOPSY SITE AND CLIP IDENTIFIED

E) BREAST, DEEP MARGIN, EXCISION:
1) LOBULAR CARCINOMA IN SITU
2) ADDITIONAL FINDINGS:
FIBROCYSTIC CHANGE WITH APOCRINE METAPLASIA
USUAL DUCAL HYPERPLASIA (UDH)
ADENOSIS
3) NEGATIVE FOR DUCTAL CARCINOMA IN SITU AND INVASIVE CARCINOMA

SYNOPTIC SUMMARY

PATHOLOGIC STAGING
PRIMARY TUMOR: pT1c
REGIONAL LYMPH NODES: pN0 (sn)
Total number of lymph nodes examined: 1
Number of lymph nodes involved: 0
Number of sentinel nodes examined: 1
Number of sentinel nodes involved: 0
Number of lymph nodes with macrometastases: 0
Number of lymph nodes with micrometastases: 0
Number of lymph nodes with isolated tumor cells: 0
DISTANT METASTASIS: Unknown

BREAST PATHOLOGIC PROGNOSTIC GROUP MARKERS
OVERALL HISTOLOGIC GRADE (Nottingham histologic grade): Grade 3
ANCILLARY STUDIES: (Previously performed by outside institution)
HER 2/neu (FISH): Negative
HER2 neu/D17Z1 ratio: 1.1
HER2 copy numbers per nucleus: 3.5
ESTROGEN RECEPTOR: Positive, >95% (strong intensity)
PROGESTERONE RECEPTOR: Negative, 0%
MIB-1 (Ki-67): Not performed

MICROSCOPIC:
TUMOR SIZE: 16.0 mm
HISTOLOGIC TYPE: Invasive carcinoma compatible with mucinous type

HISTOLOGIC GRADE:
Glandular (Acinar)/Tubular differentiation: Score 3
Nuclear pleomorphism: Score 3
Mitotic rate: Score 2
Nottingham histologic score: 8 of 9
Overall Grade: Grade 3
TUMOR FOCALITY: Single focus of invasive carcinoma
DUCTAL CARCINOMA IN-SITU (DCIS): Present
Size (extent) of DCIS: Spans 40.0 mm
Architectural Pattern(s): solid, comedo
Nuclear grade: 3
Necrosis: present
LOBULAR CARCINOMA IN SITU (LCIS): Present
TUMOR EXTENT:
Skin invasion: Not applicable
Skin satellite foci: Not applicable
Nipple: Not applicable
Skeletal muscle: Not identified
MARGINS:
Invasive carcinoma margins: All margins negative for invasive carcinoma
Distance from closest margin: 2.0 mm
Specify margin: Posterior
Anterior margin: Greater than 5.0 mm
Posterior margin: 2.0 mm
Superior margin: Greater than 5.0 mm
Inferior margin: Greater than 5.0 mm
Medial margin: Greater than 5.0 mm
Lateral margin: Greater than 5.0 mm
DCIS margins:
Anterior margin: 1.0 mm
Posterior margin: Greater than 10.0 mm (including specimen E)
Superior margin: Less than 1.0 mm
Inferior margin: Less than 1.0 mm
Medial margin: Greater than 10.0 mm
Lateral margin: Greater than 10.0 mm
TREATMENT EFFECT: No known presurgical therapy
LYMPHOVASCULAR INVASION: Not identified

MACROSCOPIC
PROCEDURE: Hookwire partial mastectomy
SPECIMEN LATERALITY: Left

ipv/1/23/2023

*** Report Electronically Signed ***
Alan Grindstaff, M.D.

Gross Description

A) Received in formalin labeled with the patient's name, Hill, Leslie, with date of birth, and "sentinel lymph node #1" is a 2 x 1.9 x 1.1 cm fatty lymph node candidate which is sectioned and entirely submitted according to Sentinel Lymph Node Protocol in A1-A4. (cassettes verified)

B) Received in formalin labeled with the patient's name, Hill, Leslie, with date of birth, and "possible sentinel lymph node #1" are two lymph node candidates, 0.8 cm and 1.3 cm, which are submitted according to Sentinel Lymph Node Protocol as follows:

SLIDE KEY:
B1) Smaller lymph node
B2) Larger lymph node
(cassettes verified)

C) Received in formalin labeled with the patient's name, Hill, Leslie, with date of birth, and "nonsentinel lymph node" are two portions of focally cauterized yellow-tan tissue, 0.5 cm and 1 cm, which are submitted whole in C1. (cassette verified)

D) Received fresh, compressed in a Dubin device labeled with the patient's name, Hill, Leslie, with date of birth, and "left hookwire partial mastectomy, wire anterior, short stitch superior 12 o'clock, long stitch lateral at 3 o'clock" is a 71-gram portion of fibrofatty breast tissue with orienting sutures and a hookwire as stated above. Using the hookwire and suture orientation, the specimen has irregular dimensions of 10 cm medial to lateral x 6.2 cm superior to inferior x 4.1 cm anterior to posterior. Anterior is inked orange, posterior is inked black, superior is inked blue, inferior is inked green, and medial and lateral are denoted by the junction of blue and green inks. The specimen is serially sectioned from medial to lateral into 18 slices. Located in slices 8-10 is a fairly well-delineated, solid, tan-pink tumor (1.5 cm medial to lateral x 1.4 cm superior to inferior x 1.2 cm anterior to posterior) which is located 3.5 cm from the medial margin, 5 cm from the lateral margin, 1.4 cm from the superior margin, 2.5 cm from the inferior margin, 1.5 cm from the anterior margin, and <0.1 cm from the posterior margin. The hookwire corresponds to the tumor and a biopsy marking clip is also present, identified within the tumor. There is a small amount of nearby hemorrhage and fat necrosis. Otherwise, the tumor is surrounded by glistening, lobulated, yellow adipose tissue. No other gross lesions are identified. Representative sections including the entire tumor are submitted in cassettes as follows:

SLIDE KEY:
D1) Medial margin, slice 1
D2) Superior half of slice 7, area before tumor
D3) Superior portion of slice 8
D4-D5) Bisected central portion of slice 8 with tumor, submitted anterior to posterior (tumor in D5)
D6) Inferior portion of slice 8
D7) Superior portion of slice 9
D8-D9) Bisected central portion of slice 9 with tumor, submitted anterior to posterior (tumor in D9)
D10) Section thinned from D9 (remainder of tumor in slice 9 with respect to posterior margin)
D11) Inferior portion of slice 9
D12-D13) Bisected central portion of slice 10 with tumor, submitted anterior to posterior (tumor in D13)
D14) Section thinned from D13 (remainder of tumor in slice 10 with respect to posterior margin
D15-D16) Bisected central portion of slice 11, area after tumor, submitted anterior to posterior
D17) Lateral margin, slice 18
(cassettes verified)

E) Received in formalin labeled with the patient's name, Hill, Leslie, with date of birth, and "deep margin, stitch marks new deep margin" is a 10-gram, 4.1 x 3.2 cm x 2.3 cm in thickness portion of fibrofatty breast tissue with a suture on one side marking this as the new margin. The new margin surface is convex and smoothly lobulated. The new margin is inked blue. The opposite non-margin aspect, which is inked black, is concave and tan-pink and slightly granular. The specimen is serially sectioned perpendicular to the long axis into 13 slices. No gross lesions are identified. The entire specimen is submitted in cassettes as follows:

SLIDE KEY:
E1) Slices 1 and 2
E2) Slice 3
E3) Slice 4
E4) Slice 5
E5) Slice 6
E6) Slice 7

E7) Slice 8
E8) Slice 9
E9) Slice 10
E10) Slice 11
E11) Slices 12 and 13
(cassettes verified)

ELH/rjc/1/18/2023

Microscopic Description

To subclassify the lesion, breast cocktail, CK5/6 and E-cadherin were performed.

Immunostain for pancytokeratin were reviewed on the sentinel lymph nodes as per protocol.

The pancytokeratin stains show no evidence of carcinoma cells in the lymph nodes. The breast cocktail stain demonstrates intact myoepithelial cells in areas of in situ carcinoma. E-cadherin is negative in LCIS. CK5/6 is positive in usual ductal hyperplasia.

The results support the diagnosis above.

Immunostain Summary:
Pancytokeratin: A1-A4, B1-B2
Breast cocktail: D17
E-cadherin: E6-7
CK5/6: E2

In some tests, analyte specific reagents (ASRs) are used. In the case of an ASR, this test was developed and its performance characteristics determined by this laboratory. It has not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA '88) as qualified to perform high complexity clinical laboratory testing.